Practical no. 3

Aim – Interface TM1637 with Raspberry Pi and Display current time.

To display small amount of data with Raspberry Pi, we can use 4 digit 7-segment Display. 7 Segment Display has seven segments in it and each segment has one LED inside it to display the numbers by lighting up the corresponding segments.

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. 4 digit 7 Segment Display
3. Jumper wires (Female to Female)

Here, I am using **4 digits-7 segments LED** display with TM1637 controller





- **Software Requirements**
  1. Raspbian Stretch OS

1. **Connect your 4 digit 7 segment display with Raspberry Pi's GPIO Pins.**

TM1637 Board Pin	Function	RPI Physical Pin	Raspberry Function
GND	Ground	14	GND
VCC	+ 5V Power	4	5V
DI0	Data In	18	GPIO 24
CLK	Clock	16	GPIO 23

Open terminal –
pip3 show tm1637
pip3 install python-tm1637
pip3 install python-tm1637

Code: -

```
import time
import datetime
import tm1637
# CLK -> GPIO23 (Pin 16)
# DIO -> GPIO24 (Pin 18)
# Create the TM1637 display
Display = tm1637.TM1637(clk=23, dio=24)
# Set brightness
print("4 Digit Display - Digital Clock")
# Start with colon ON
colon = True
while True:
    # Get the current time
    now = datetime.datetime.now()
    # Get hour and minute
    hour = now.hour
    minute = now.minute
    # Display hour and minute
    # sep=True displays the colon
    Display.numbers(hour, minute, sep=colon)
    # Change colon state
    colon = not colon
    # Wait for one second
    time.sleep(1)
```

O/P-

